

2

Machines & Models

14 October 2025

Prof. Dr. Sebastian Wild

Learning Outcomes

Unit 2: *Machines & Models*

1. Understand the difference between empirical *running time* and algorithm *analysis*.
2. Understand *worst / best / average case* models for input data.
3. Know the *RAM machine* model.
4. Know the definitions of *asymptotic notation* (Big-Oh classes and relatives).
5. Understand the reasons to make *asymptotic approximations*.
6. Be able to *analyze* simple *algorithms*.

Outline

2 Machines & Models

- 2.1 Algorithm analysis
- 2.2 The RAM Model
- 2.3 Asymptotics & Big-Oh
- 2.4 Asymptotic Pitfalls
- 2.5 Teaser: The Maximum subarray problem
- 2.6 Teaser: Algorithm Design in Action

What is an algorithm?

An algorithm is a sequence of instructions.

think: recipe

e. g. Python script

More precisely:

1. mechanically executable
~> no "common sense" needed
2. finite description $\neq$ finite computation!
3. solves a *problem*, i. e., a class of problem instances

$x + y$, not only $17 + 4$

► input-processing-output abstraction



Typical example: *bubblesort*

~> not a specific program
but the underlying idea

What is a data structure?

A data structure is

1. a rule for **encoding data**
(in computer memory), plus
2. **algorithms** to work with it
(queries, updates, etc.)

typical example: *binary search tree*



2.1 Algorithm analysis

Good algorithms

Our goal: Find good (best?) algorithms and data structures for a task.

Good “usually” means

- ▶ fast running *time*
- ▶ moderate memory *space* usage

↙ can be complicated in distributed systems

Algorithm analysis is a way to

- ▶ compare different algorithms,
- ▶ predict their performance in an application

Running time experiments

Why not simply run and time it?

- ▶ results only apply to
 - ▶ single *test* machine
 - ▶ tested inputs
 - ▶ tested implementation
 - ▶ ...

$\neq$ *universal truths*

- ▶ instead: consider and analyze algorithms on an abstract machine

~> provable statements for model

~> testable model hypotheses



survives Pentium 4

~> Need precise model of machine (costs), input data and algorithms.

Data Models

Algorithm analysis typically uses one of the following simple data models:

- ▶ **worst-case performance:**
consider the *worst* of all inputs as our cost metric
- ▶ **best-case performance:**
consider the *best* of all inputs as our cost metric
- ▶ **average-case performance:**
consider the average/expectation of a *random* input as our cost metric

Usually, we apply the above for *inputs of same size n* .

↪ performance is only a **function of n** .

2.2 The RAM Model

Clicker Question



What is the cost of *adding* two d -digit integers?
(For example, for $d = 5$, what is $45\,235 + 91\,342$?)

- A** constant time
- B** logarithmic in d
- C** proportional to d
- D** quadratic in d
- E** no idea what you are talking about



→ sli.do/cs566

Clicker Question



What is the cost of *adding* two d -digit integers?
(For example, for $d = 5$, what is $45\,235 + 91\,342$?)

- ☒ **A** constant time ✓
- ☐ **B** ~~logarithmic in d~~
- ☒ **C** proportional to d ✓
- ☐ **D** ~~quadratic in d~~
- ☒ **E** no idea what you are talking about ✓



→ sli.do/cs566

Machine models

The machine model decides

- ▶ what algorithms are possible
- ▶ how they are described (= programming language)
- ▶ what an execution *costs*

Goal: Machine models should be
detailed and powerful enough to reflect actual machines,
abstract enough to unify architectures,
simple enough to analyze.

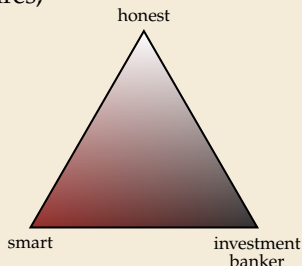
Machine models

The machine model decides

- ▶ what algorithms are possible
- ▶ how they are described (= programming language)
- ▶ what an execution *costs*

Goal: Machine models should be
detailed and powerful enough to reflect actual machines,
abstract enough to unify architectures,
simple enough to analyze.

~> usually some compromise is needed



Random Access Machines

Random access machine (RAM)

more detail in §2.2 of *Sequential and Parallel Algorithms and Data Structures*
by Sanders, Mehlhorn, Dietzfelbinger, Dementiev

- ▶ unlimited *memory* $\text{MEM}[0], \text{MEM}[1], \text{MEM}[2], \dots$
- ▶ fixed number of *registers* $R_1, \dots, R_r$ (say $r = 100$)

Random Access Machines

Random access machine (RAM)

more detail in §2.2 of *Sequential and Parallel Algorithms and Data Structures*
by Sanders, Mehlhorn, Dietzfelbinger, Dementiev

- ▶ unlimited *memory* $\text{MEM}[0], \text{MEM}[1], \text{MEM}[2], \dots$
- ▶ fixed number of *registers* $R_1, \dots, R_r$ (say $r = 100$)
- ▶ memory cells $\text{MEM}[i]$ and registers R_i store w -bit integers, i. e., numbers in $[0..2^w - 1]$
 w is the word width/size; typically $w \propto \lg n \Rightarrow 2^w \approx n$

Random Access Machines

Random access machine (RAM)

more detail in §2.2 of *Sequential and Parallel Algorithms and Data Structures*
by Sanders, Mehlhorn, Dietzfelbinger, Dementiev

- ▶ unlimited *memory* $\text{MEM}[0], \text{MEM}[1], \text{MEM}[2], \dots$
- ▶ fixed number of *registers* $R_1, \dots, R_r$ (say $r = 100$)
- ▶ memory cells $\text{MEM}[i]$ and registers R_i store w -bit integers, i. e., numbers in $[0..2^w - 1]$
 w is the word width/size; typically $w \propto \lg n \implies 2^w \approx n$
- ▶ Instructions:
 - ▶ load & store: $R_i := \text{MEM}[R_j]$ $\text{MEM}[R_j] := R_i$
 - ▶ operations on registers: $R_k := R_i + R_j$ (arithmetic is *modulo* 2^w !)
also $R_i - R_j$, $R_i \cdot R_j$, $R_i \text{ div } R_j$, $R_i \bmod R_j$
C-style operations (bitwise and/or/xor, left/right shift)
 - ▶ conditional and unconditional jumps
- ▶ cost: number of executed instructions

we will see further models later

$\rightsquigarrow$ The RAM is the standard model for sequential computation.

RAM-Program Example

Example RAM program

```
1 // Assume:  $R_1$  stores number  $N$ 
2 // Assume:  $\text{MEM}[0..N)$  contains list of  $N$  numbers
3  $R_2 := R_1$ ;
4  $R_3 := R_1 - 2$ ;
5  $R_4 := \text{MEM}[R_3]$ ;
6  $R_5 := R_3 + 1$ ;
7  $R_6 := \text{MEM}[R_5]$ ;
8 if ( $R_4 \leq R_6$ ) goto line 11;
9  $\text{MEM}[R_3] := R_6$ ;
10  $\text{MEM}[R_5] := R_4$ ;
11  $R_3 := R_3 - 1$ ;
12 if ( $R_3 \geq 0$ ) goto line 5;
13  $R_2 := R_2 - 1$ ;
14 if ( $R_2 > 0$ ) goto line 4;
15 // Done:
```

Clicker Question



What algorithm does the RAM program on the previous slide implement?



→ *sli.do/cs566*

RAM-Program Example

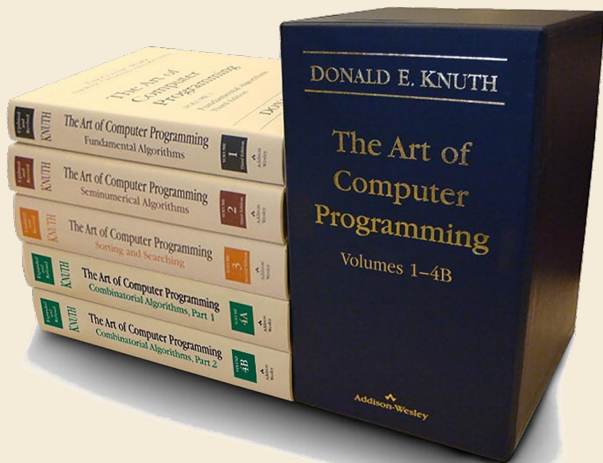
Example RAM program

```
1 // Assume:  $R_1$  stores number  $N$ 
2 // Assume:  $\text{MEM}[0..N)$  contains list of  $N$  numbers
3  $R_2 := R_1$ ;
4  $R_3 := R_1 - 2$ ;
5  $R_4 := \text{MEM}[R_3]$ ;
6  $R_5 := R_3 + 1$ ;
7  $R_6 := \text{MEM}[R_5]$ ;
8 if ( $R_4 \leq R_6$ ) goto line 11;
9  $\text{MEM}[R_3] := R_6$ ;
10  $\text{MEM}[R_5] := R_4$ ;
11  $R_3 := R_3 - 1$ ;
12 if ( $R_3 \geq 0$ ) goto line 5;
13  $R_2 := R_2 - 1$ ;
14 if ( $R_2 > 0$ ) goto line 4;
15 // Done:  $\text{MEM}[0..N)$  sorted
```

RAM-Program Example

Example RAM program

```
1 // Assume:  $R_1$  stores number  $N$ 
2 // Assume:  $\text{MEM}[0..N)$  contains list of  $N$  numbers
3  $R_2 := R_1$ ;
4  $R_3 := R_1 - 2$ ;
5  $R_4 := \text{MEM}[R_3]$ ;
6  $R_5 := R_3 + 1$ ;
7  $R_6 := \text{MEM}[R_5]$ ;
8 if ( $R_4 \leq R_6$ ) goto line 11;
9  $\text{MEM}[R_3] := R_6$ ;
10  $\text{MEM}[R_5] := R_4$ ;
11  $R_3 := R_3 - 1$ ;
12 if ( $R_3 \geq 0$ ) goto line 5;
13  $R_2 := R_2 - 1$ ;
14 if ( $R_2 > 0$ ) goto line 4;
15 // Done:  $\text{MEM}[0..N)$  sorted
```



RAM-Program Example

Example RAM program

```
1 // Assume:  $R_1$  stores number  $N$ 
2 // Assume:  $\text{MEM}[0..N]$  contains list of  $N$  number
3  $R_2 := R_1$ ;
4  $R_3 := R_1 - 2$ ;
5  $R_4 := \text{MEM}[R_3]$ ;
6  $R_5 := R_3 + 1$ ;
7  $R_6 := \text{MEM}[R_5]$ ;
8 if ( $R_4 \leq R_6$ ) goto line 11;
9  $\text{MEM}[R_3] := R_6$ ;
10  $\text{MEM}[R_5] := R_4$ ;
11  $R_3 := R_3 - 1$ ;
12 if ( $R_3 \geq 0$ ) goto line 5;
13  $R_2 := R_2 - 1$ ;
14 if ( $R_2 > 0$ ) goto line 4;
15 // Done:  $\text{MEM}[0..N]$  sorted
```

5.2.2

SORTING BY EXCHANGING 107

they need not be examined on subsequent passes. Horizontal lines in Fig. 14 show the progress of the sorting from this standpoint; notice, for example, that five more elements are known to be in final position as a result of Pass 4. On the final pass, no exchanges are performed at all. With these observations we are ready to formulate the algorithm.

Algorithm B (*Bubble sort*). Records $R_1, \dots, R_N$ are rearranged in place; after sorting is complete their keys will be in order, $K_1 \leq \dots \leq K_N$.

B1. [Initialize **BOUND**.] Set $\text{BOUND} \leftarrow N$. (**BOUND** is the highest index for which the record is not known to be in its final position; thus we are indicating that nothing is known at this point.)

B2. [Loop on j .] Set $t \leftarrow 0$. Perform step B3 for $j = 1, 2, \dots, \text{BOUND} - 1$, and then go to step B4. (If $\text{BOUND} = 1$, this means go directly to B4.)

B3. [Compare/exchange $R_j : R_{j+1}$.] If $K_j > K_{j+1}$, interchange $R_j \leftrightarrow R_{j+1}$ and set $t \leftarrow j$.

B4. [Any exchanges?] If $t = 0$, terminate the algorithm. Otherwise set $\text{BOUND} \leftarrow t$ and return to step B2. ■

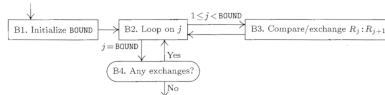


Fig. 15. Flow chart for bubble sorting.

Program B (*Bubble sort*). As in previous MIX programs of this chapter, we assume that the items to be sorted are in locations $\text{INPUT}+1$ through $\text{INPUT}+N$. $\text{r11} \equiv t$; $\text{r12} \equiv j$.

01	START	ENT1	N	1	B1. Initialize BOUND .	$t \leftarrow N$.
02	1H	ST1	BOUND (1:2)	A	BOUND $\leftarrow t$.	
03		ENT2	1	A	B2. Loop on j .	$j \leftarrow 1$.
04		ENT1	0	A		$t \leftarrow 0$.
05		JMP	BOUND	A	Exit if $j \geq \text{BOUND}$.	
06	3H	LDA	INPUT ,2	C	B3. Compare/exchange $R_j : R_{j+1}$.	
07		CMPA	INPUT +1,2	C		
08		JLE	2F	C	No exchange if $K_j \leq K_{j+1}$.	
09		LDX	INPUT +1,2	B	R_{j+1}	
10		STX	INPUT ,2	B	$\rightarrow R_j$.	
11		STA	INPUT +1,2	B	(old R_j) $\rightarrow R_{j+1}$.	
12		ENT1	0,2	B	$t \leftarrow j$.	
13	2H	INC2	1	C	$j \leftarrow j + 1$.	
14	BOUND	ENTX	$\rightarrow$,2	A + C	$\text{rX} \leftarrow j - \text{BOUND}$.	[Instruction modified]
15		JXN	3B	A + C	Do step B3 for $1 \leq j < \text{BOUND}$.	
16	4H	J1P	1B	A	B4. Any exchanges? To B2 if $t > 0$.	■

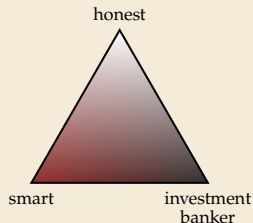
Pseudocode

- ▶ Programs for the random-access machine are very low level and detailed
≈ assembly/machine language

Typical simplifications when describing and analyzing algorithms:

- ▶ more abstract *pseudocode* ← code that humans understand (easily)
 - ▶ control flow using **if**, **for**, **while**, etc.
 - ▶ variable names instead of fixed registers and memory cells
 - ▶ memory management (more below)
- ▶ count dominant *elementary operations* (e. g. memory accesses) instead of all RAM instructions

In both cases: We *can* go to full detail where needed/desired.



Pseudocode – Example

RAM-Program

```
1 // Bubblesort
2 // Assume:  $R_1$  stores number  $N$ 
3 // Assume:  $\text{MEM}[0..N)$  contains list of  $N$  numbers
4  $R_2 := R_1$ ;
5  $R_3 := R_1 - 2$ ;
6  $R_4 := \text{MEM}[R_3]$ ;
7  $R_5 := R_3 + 1$ ;
8  $R_6 := \text{MEM}[R_5]$ ;
9 if ( $R_4 \leq R_6$ ) goto line 12;
10  $\text{MEM}[R_3] := R_6$ ;
11  $\text{MEM}[R_5] := R_4$ ;
12  $R_3 := R_3 - 1$ ;
13 if ( $R_3 \geq 0$ ) goto line 6;
14  $R_2 := R_2 - 1$ ;
15 if ( $R_2 > 0$ ) goto line 5;
16 // Done:  $\text{MEM}[0..N)$  sorted
```

Pseudocode Algorithm

```
1 procedure bubblesort( $A[0..N)$ ):
2   for  $i := N, N - 1, \dots, 1$ 
3     for  $j := N - 2, N - 3, \dots, 0$ 
4       if  $A[j] > A[j + 1]$ :
5         Swap  $A[j]$  and  $A[j + 1]$ 
6       end if
8     end for
```

⇒ much more **readable**

- ▶ closer to modern high-level programming languages
- ▶ **but:** only allow primitive operations that correspond to $O(1)$ RAM instructions
 - ⇒ analysis

Memory management & Pointers

- ▶ A random-access machine is a bit like a bare CPU . . . without any operating system
 - ↪ cumbersome to use
- ▶ All high-level programming languages / operating systems add *memory management*:
 - ▶ Instruction to *allocate* a contiguous piece of memory of a given size (like `malloc`).
 - ▶ used to allocate a new array (of a fixed size) or
 - ▶ a new object/record (with a known list of instance variables)
 - ▶ There's a similar instruction to free allocated memory again or an automated garbage collector.
 - ↪ A *pointer* is a memory address (i. e., the *i* of `MEM[i]`).
- ▶ Support for procedures (a.k.a. functions, methods) calls including recursive calls
 - ▶ (this internally requires maintaining call stack)



We will mostly ignore *how* all this works here.

2.3 Asymptotics & Big-Oh

Clicker Question

What is the correct way to complete the equation?

$$8n + \frac{1}{2}n^2 + 1024 = \square$$



- A** $O(1)$
- B** $O(n)$
- C** $O(n \log(n))$
- D** $O(n^2)$
- E** I don't know $O(\cdot)$



→ sli.do/cs566

Clicker Question

What is the correct way to complete the equation?

$$8n + \frac{1}{2}n^2 + 1024 = \square$$



- A** ~~$O(1)$~~
- B** ~~$O(n)$~~
- C** ~~$O(n \log(n))$~~
- D** $O(n^2)$ ✓
- E** ~~I don't know $O(\cdot)$~~



→ sli.do/cs566

Why asymptotics?

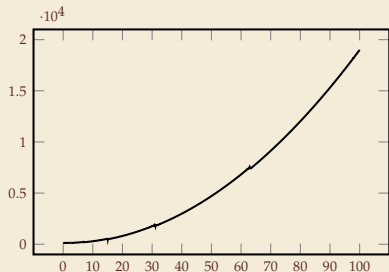
Algorithm analysis focuses on (the limiting behavior for infinitely) **large** inputs.

- ▶ abstracts from unnecessary detail
- ▶ simplifies analysis
- ▶ often necessary for sensible comparison

Asymptotics = approximation around ∞

Example: Consider a function $f(n)$ given by

$$2n^2 - 3n\lfloor \log_2(n+1) \rfloor + 7n - 3\lfloor \log_2(n+1) \rfloor + 120$$



Why asymptotics?

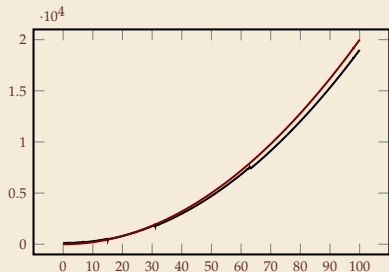
Algorithm analysis focuses on (the limiting behavior for infinitely) **large** inputs.

- ▶ abstracts from unnecessary detail
- ▶ simplifies analysis
- ▶ often necessary for sensible comparison

Asymptotics = approximation around ∞

Example: Consider a function $f(n)$ given by

$$2n^2 - 3n\lfloor \log_2(n+1) \rfloor + 7n - 3\lfloor \log_2(n+1) \rfloor + 120 \sim 2n^2$$



Asymptotic tools – Formal & definitive definition

► “Tilde Notation”:

$$f(n) \sim g(n) \quad \overset{\text{if, and only if}}{\text{iff}} \quad \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 1$$

„ f and g are *asymptotically equivalent*”

Asymptotic tools – Formal & definitive definition

► “Tilde Notation”: $f(n) \sim g(n)$ ^{if, and only if} iff $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 1$
„ f and g are asymptotically equivalent”

► “Big-Oh Notation”: $f(n) \in O(g(n))$ ^{also write ‘=’ instead} iff $\left| \frac{f(n)}{g(n)} \right|$ is bounded for $n \geq n_0$

^{need supremum since limit might not exist!} iff $\lim_{n \rightarrow \infty} \sup \left| \frac{f(n)}{g(n)} \right| < \infty$

Variants: “Big-Omega”

► $f(n) \in \Omega(g(n))$ ^{“Big-Omega”} iff $g(n) \in O(f(n))$

► $f(n) \in \Theta(g(n))$ ^{“Big-Theta”} iff $f(n) \in O(g(n))$ **and** $f(n) \in \Omega(g(n))$

Asymptotic tools – Formal & definitive definition

- “Tilde Notation”: $f(n) \sim g(n)$ ^{if, and only if} $\iff \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 1$
„ f and g are asymptotically equivalent”

- “Big-Oh Notation”: $f(n) \in O(g(n))$ ^{also write ‘=’ instead} $\iff \left| \frac{f(n)}{g(n)} \right|$ is bounded for $n \geq n_0$

^{need supremum since limit might not exist!}
 $\iff \lim_{n \rightarrow \infty} \sup \left| \frac{f(n)}{g(n)} \right| < \infty$

Variants: “Big-Omega”

- $f(n) \in \Omega(g(n))$ $\iff g(n) \in O(f(n))$
► $f(n) \in \Theta(g(n))$ $\iff f(n) \in O(g(n))$ **and** $f(n) \in \Omega(g(n))$

“Big-Theta”

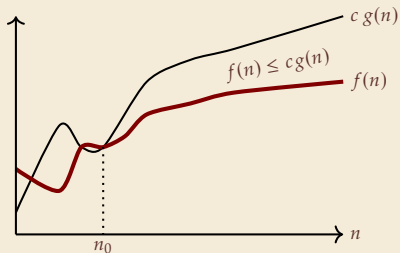
- “Little-Oh Notation”: $f(n) \in o(g(n))$ $\iff \lim_{n \rightarrow \infty} \left| \frac{f(n)}{g(n)} \right| = 0$

similarly: $f(n) \in \omega(g(n))$ if $\lim = \infty$

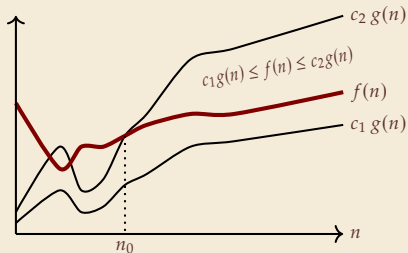
(Benefit of this definition: Works for any $f, g : \mathbb{R} \rightarrow \mathbb{R}$ and is easy to generalize to limits other than $n \rightarrow \infty$)

Asymptotic tools – Intuition

- $f(n) = O(g(n))$: $f(n)$ is **at most** $g(n)$
up to constant factors and
for sufficiently large n



- $f(n) = \Theta(g(n))$: $f(n)$ is **equal to** $g(n)$
up to constant factors and
for sufficiently large n



Plots can be misleading!

Example ↗

Clicker Question



Assume $f(n) \in O(g(n))$. What can we say about $g(n)$?

- A** $g(n) = O(f(n))$
- B** $g(n) = \Omega(f(n))$
- C** $g(n) = \Theta(f(n))$
- D** Nothing (it depends on f and g)



→ sli.do/cs566

Clicker Question



Assume $f(n) \in O(g(n))$. What can we say about $g(n)$?

- A** ~~$g(n) = O(f(n))$~~
- B** $g(n) = \Omega(f(n))$ ✓
- C** ~~$g(n) = \Theta(f(n))$~~
- D** ~~Nothing (it depends on f and g)~~



→ sli.do/cs566

Clicker Question



Assume $f(n) \in O(g(n))$. What can we say about $g(n)$?

- A** ~~$g(n) = O(f(n))$~~
- B** $g(n) = \Omega(f(n))$ ✓ (if $f(n) \neq 0$)
- C** ~~$g(n) = \Theta(f(n))$~~
- D** Nothing (it depends on f and g) ✓



→ sli.do/cs566

Asymptotics – Example 1

Basic examples:

- ▶ $20n^3 + 10n \ln(n) + 5 \sim 20n^3 = \Theta(n^3)$
 $= O(n^3)$
 $= \Omega(n^3)$
- ▶ $3 \lg(n^2) + \lg(\lg(n)) = \Theta(\log n)$
- ▶ $10^{100} = O(1)$

$$\lim_{n \rightarrow \infty} \left| \frac{20n^3 + 10n \ln n + 5}{20n^3} \right| =$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{20n^3}}{\cancel{20n^3}} 1 + \lim_{n \rightarrow \infty} \underbrace{\frac{10n \ln n}{20n^3}}_{\leq \frac{1}{2n}} + \lim_{n \rightarrow \infty} \underbrace{\frac{5}{20n^3}}_{=0}$$

$$= 1 < \infty$$

$$\lim_{n \rightarrow \infty} \frac{20n^3}{20n^3 + 10n \ln n + 5} = \lim_{n \rightarrow \infty} \frac{\cancel{20n^3}}{\cancel{20n^3} \left(1 + \underbrace{\frac{10n \ln n}{20n^3}}_{\rightarrow 0} + \underbrace{\frac{5}{20n^3}}_{\rightarrow 0} \right)} = 1$$

Use *wolfram alpha* to compute/check limits, but also practice it with pen and paper!

Clicker Question



Is $(\sin(n) + 2)n^2 = \Theta(n^2)$?

A

Yes

B

No



→ *sli.do/cs566*

Clicker Question



Is $(\sin(n) + 2)n^2 = \Theta(n^2)$?

A Yes ✓

$\lim_{n \rightarrow \infty} \left| \frac{(\sin(n) + 2) \cancel{n^2}}{\cancel{n^2}} \right|$ ex. nicht!

B ~~No~~

aber:

$$1 \leq \left| \frac{(\sin(n) + 2) \cancel{n^2}}{\cancel{n^2}} \right| \leq 3$$

$$\Rightarrow (\sin(n) + 2)n^2 = O(n^2)$$



→ sli.do/cs566

Asymptotics – Basic facts

Rules to work with Big-Oh classes:

- ▶ $f = \Theta(f)$ (reflexivity)
- ▶ $f = \Theta(g) \wedge g = \Theta(h) \implies f = \Theta(h)$
- ▶ $c \cdot f(n) = \Theta(f(n))$ for constant $c \neq 0$
- ▶ $f \sim g \iff f = g \cdot \underline{1 \pm o(1)}$
- ▶ $\Theta(f) \cdot \Theta(g) = \Theta(f \cdot g)$
- ▶ $\Theta(f) + \Theta(g) = \Theta(f + g) = \Theta(\max\{f, g\})$ largest summand determines Θ -class

Asymptotics – Frequently encountered classes

Frequently used orders of growth:

- ▶ constant $\Theta(1)$
- ▶ logarithmic $\Theta(\log n)$ Note: $a, b > 0$ constants $\rightsquigarrow \Theta(\log_a(n)) = \Theta(\log_b(n))$
- ▶ linear $\Theta(n)$
- ▶ linearithmic $\Theta(n \log n)$
- ▶ quadratic $\Theta(n^2)$
- ▶ cubic $\Theta(n^3)$
- ▶ polynomial $O(n^c)$ for some constant c
- ▶ exponential $O(c^n)$ for some constant $c > 1$ Note: $a > b > 0$ constants $\rightsquigarrow b^n = o(a^n)$

2.4 Asymptotic Pitfalls

Asymptotics – Example 2

Square-and-multiply algorithm

for computing x^m with $m \in \mathbb{N}$

Inputs:

- ▶ m as binary number (array of bits)
- ▶ $n = \text{\#bits in } m$
- ▶ x a floating-point number

```
1 def pow(x, m):
2     # compute binary representation of exponent
3     exponent_bits = bin(m)[2:]
4     result = 1
5     for bit in exponent_bits:
6         result *= result    // n executions
7         if bit == '1':
8             result *= x    // 2     # 1s
9     return result
```

- ▶ Cost: $C = \text{\#multiplications}$
- ▶ $C = n$ (line 6) + $\text{\#one-bits in binary representation of } m$ (line 8)

$$\rightsquigarrow n \leq C \leq 2n$$

Clicker Question



We showed $n \leq C(n) \leq 2n$; what is the most precise asymptotic approximation for $C(n)$ that we can make?

Write e. g. $O(n^2)$ for $O(n^2)$ or $\Theta(\sqrt{n})$ for $\Theta(\sqrt{n})$.



→ sli.do/cs566

Asymptotics – Example 2

Square-and-multiply algorithm

for computing x^m with $m \in \mathbb{N}$

Inputs:

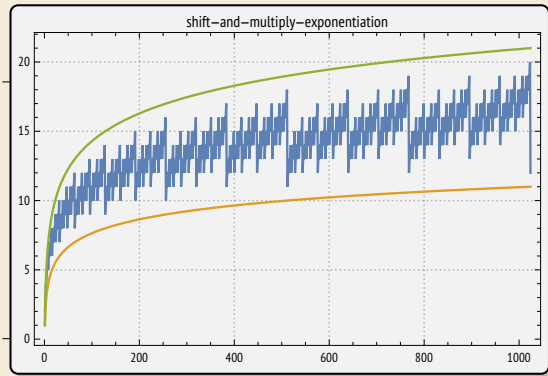
- ▶ m as binary number (array of bits)
- ▶ $n = \text{\#bits in } m$
- ▶ x a floating-point number

▶ Cost: $C = \text{\#multiplications}$

▶ $C = n$ (line 6) + $\text{\#one-bits in binary representation of } m$ (line 8)

$$\rightsquigarrow n \leq C \leq 2n$$

$$\rightsquigarrow C = \Theta(n) = \Theta(\log m)$$



Often, you can pretend Θ is “like $\sim$ with an unknown constant”
but in this case, *no such constant exists!*

Asymptotics with several variables

- ▶ **Example:** Algorithms on graphs with n vertices and m edges.
 - ▶ want to say: Algorithm A takes time $\Theta(n + m)$.
 - ▶ But what does that even mean, formally?!

Schlechte Alternative

$$\exists c \exists n_0 \exists m_0 \forall n \geq n_0 \forall m \geq m_0 \quad f(n, m) \leq c \cdot g(n, m)$$

Asymptotics with several variables

► **Example:** Algorithms on graphs with n vertices and m edges.

► want to say: Algorithm A takes time $\Theta(n + m)$.

► But what does that even mean, formally?!

⚠ Inconsistent and incompatible definitions used in the literature!

► **Here:**

► (implicitly) always have a single “*main*” variable n : with $n \rightarrow \infty$

► all other variables are *functions* of n : $m = m(n)$

► must make *conditions* on functions explicit: $m(n) \in \Omega(n)$ and $m(n) \in O(n^2)$.

↪ Can make statements like

$$O(n + m) \subseteq O(nm) \quad (n \rightarrow \infty, m \in \Omega(1))$$

$$\frac{n + m(n)}{nm(n)} = \frac{\cancel{n}}{m(n)} + \frac{\cancel{m(n)}}{nm(n)}$$

2.5 Teaser: The Maximum subarray problem

Bring on the puzzles!

Time for a concrete example of algorithm design!

- ▶ we will illustrate the algorithm design process on a “toy problem”
- ▶ clean abstract problem, but nontrivial to solve!

Bring on the puzzles!

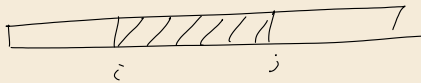
Time for a concrete example of algorithm design!

- ▶ we will illustrate the algorithm design process on a “toy problem”
- ▶ clean abstract problem, but nontrivial to solve!

Maximum (sum) subarray problem

- ▶ **Given:** $A[0..n)$ with $A[i] \in \mathbb{Z}$ for $0 \leq i < n$.
- ▶ Abbreviate $s(i, j) := \sum_{k=i}^{j-1} A[k]$
- ▶ **Goal:** Compute $\textcircled{s} := \max\{s(i, j) : 0 \leq i \leq j \leq n\}$
(and a pair (i, j) with $s = s(i, j)$.)

will ignore that here; easy to modify algorithms



$[i..j)$

$$s(i, i) = 0$$

Bring on the puzzles!

Time for a concrete example of algorithm design!

- ▶ we will illustrate the algorithm design process on a “toy problem”
- ▶ clean abstract problem, but nontrivial to solve!

Maximum (sum) subarray problem

- ▶ **Given:** $A[0..n)$ with $A[i] \in \mathbb{Z}$ for $0 \leq i < n$.
- ▶ Abbreviate $s(i, j) := \sum_{k=i}^{j-1} A[k]$
- ▶ **Goal:** Compute $s := \max\{s(i, j) : 0 \leq i \leq j \leq n\}$
and a pair (i, j) with $s = s(i, j)$.

will ignore that here; easy to modify algorithms

Applications:

- ▶ largest gain of a stock
 $A[i]$ price change on day i
- ▶ signal detection in
biological sequence
analysis
- ▶ 2D generalization used in
image analysis

Bring on the puzzles!

Time for a concrete example of algorithm design!

- ▶ we will illustrate the algorithm design process on a “toy problem”
- ▶ clean abstract problem, but nontrivial to solve!

Maximum (sum) subarray problem

- ▶ **Given:** $A[0..n]$ with $A[i] \in \mathbb{Z}$ for $0 \leq i < n$.
- ▶ Abbreviate $s(i, j) := \sum_{k=i}^{j-1} A[k]$
- ▶ **Goal:** Compute $s := \max\{s(i, j) : 0 \leq i \leq j \leq n\}$
and a pair (i, j) with $s = s(i, j)$.

will ignore that here; easy to modify algorithms

Modeling decisions:

- ▶ input size: # numbers n
 - ▶ assume all integers (and sums) fit in $O(1)$ words
- $\rightsquigarrow$ count # additions as elementary operation

Applications:

- ▶ largest gain of a stock
 $A[i]$ price change on day i
- ▶ signal detection in
biological sequence
analysis
- ▶ 2D generalization used in
image analysis

Clicker Question



What do you think is the Θ -class of the running time of the fastest algorithm for the maximal sum subarray problem?



→ *sli.do/cs566*

Template for Describing an Algorithm

1. 💡 **Algorithmic Idea**

Abstract idea that makes the algorithm work (prose)
(an expert could fill in the rest from here)

2. </> **Pseudocode**

structured description of procedure including edge cases
should be unambiguous and close to real code

3. © **Correctness proof**

argument why the correct result is computed
often uses induction and invariants

4. 🏔️ **Algorithm analysis**

analysis of the efficiency of the algorithm
usually want Θ -class of worst-case running time
where interesting, also space usage

Brute force approach

- ▶ Let's start with the simplest thinkable solution

1. 💡 Algorithmic Idea

try all contiguous subarrays $A[i..j]$

Maximal subarray problem

- ▶ **Given:** $A[0..n)$ with $A[i] \in \mathbb{Z}$ for $0 \leq i < n$.
- ▶ Abbreviate $s(i, j) := \sum_{k=i}^{j-1} A[k]$
- ▶ **Goal:** Compute $s := \max\{s(i, j) : 0 \leq i \leq j \leq n\}$
and a pair (i, j) with $s = s(i, j)$.

Brute force approach

- Let's start with the simplest thinkable solution

1. 💡 Algorithmic Idea

try all contiguous subarrays $A[i..j]$

2. </> Pseudocode

```
1  s = 0
2  for i = 0, ..., n - 1
3      for j = i, ..., n
4          t = 0
5          for k = i, ..., j - 1
6              t = t + A[k]
7          end for
8          if t > s then s := t
9      end for
10 end for
```

Maximal subarray problem

► **Given:** $A[0..n]$ with $A[i] \in \mathbb{Z}$ for $0 \leq i < n$.

► Abbreviate $s(i, j) := \sum_{k=i}^{j-1} A[k]$

► **Goal:** Compute $s := \max\{s(i, j) : 0 \leq i \leq j \leq n\}$
and a pair (i, j) with $s = s(i, j)$.

Brute force approach

- Let's start with the simplest thinkable solution

1. 💡 Algorithmic Idea

try all contiguous subarrays $A[i..j]$

2. </> Pseudocode

```
1  s = 0
2  for i = 0, ..., n - 1
3      for j = i, ..., n
4          t = 0
5          for k = i, ..., j - 1
6              t = t + A[k]
7          end for
8          if t > s then s := t
9      end for
10 end for
```

3. ☉ Correctness proof

direct by definition of s

Maximal subarray problem

► **Given:** $A[0..n]$ with $A[i] \in \mathbb{Z}$ for $0 \leq i < n$.

► Abbreviate $s(i, j) := \sum_{k=i}^{j-1} A[k]$

► **Goal:** Compute $s := \max\{s(i, j) : 0 \leq i \leq j \leq n\}$
and a pair (i, j) with $s = s(i, j)$.

Brute force approach

- Let's start with the simplest thinkable solution

1. 💡 Algorithmic Idea

try all contiguous subarrays $A[i..j]$

2. </> Pseudocode

```
1 s = 0
2 for i = 0, ..., n - 1
3     for j = i, ..., n
4         t = 0
5         for k = i, ..., j - 1
6             t = t + A[k]
7         end for
8         if t > s then s := t
9     end for
10 end for
```

3. ☉ Correctness proof

direct by definition of s

Maximal subarray problem

- **Given:** $A[0..n]$ with $A[i] \in \mathbb{Z}$ for $0 \leq i < n$.
- Abbreviate $s(i, j) := \sum_{k=i}^{j-1} A[k]$
- **Goal:** Compute $s := \max\{s(i, j) : 0 \leq i \leq j \leq n\}$ and a pair (i, j) with $s = s(i, j)$.

4. 🏔️ Algorithm analysis

additions

Brute force approach

- Let's start with the simplest thinkable solution

1. 💡 Algorithmic Idea

try all contiguous subarrays $A[i..j]$

2. </> Pseudocode

```
1 s = 0
2 for i = 0, ..., n - 1
3   for j = i, ..., n
4     t = 0
5     for k = i, ..., j - 1
6       t = t + A[k]
7     end for
8     if t > s then s := t
9   end for
10 end for
```

Maximal subarray problem

► **Given:** $A[0..n]$ with $A[i] \in \mathbb{Z}$ for $0 \leq i < n$.

► Abbreviate $s(i, j) := \sum_{k=i}^{j-1} A[k]$

► **Goal:** Compute $s := \max\{s(i, j) : 0 \leq i \leq j \leq n\}$ and a pair (i, j) with $s = s(i, j)$.

4. 🏔️ Algorithm analysis

additions $= j - i - 1 + 1 = j - i$

$$= \sum_{i=0}^{n-1} \sum_{j=i}^n \sum_{k=i}^{j-1} 1$$

3. ☉ Correctness proof

direct by definition of s

Brute force approach

- Let's start with the simplest thinkable solution

1. 💡 Algorithmic Idea

try all contiguous subarrays $A[i..j]$

2. </> Pseudocode

```
1 s = 0
2 for i = 0, ..., n - 1
3     for j = i, ..., n
4         t = 0
5         for k = i, ..., j - 1
6             t = t + A[k]
7         end for
8         if t > s then s := t
9     end for
10 end for
```

3. ☉ Correctness proof

direct by definition of s

Maximal subarray problem

- **Given:** $A[0..n]$ with $A[i] \in \mathbb{Z}$ for $0 \leq i < n$.
- Abbreviate $s(i, j) := \sum_{k=i}^{j-1} A[k]$
- **Goal:** Compute $s := \max\{s(i, j) : 0 \leq i \leq j \leq n\}$ and a pair (i, j) with $s = s(i, j)$.

4. 🏔️ Algorithm analysis

additions

$$= \sum_{i=0}^{n-1} \sum_{j=i}^n \sum_{k=i}^{j-1} 1 = \sum_{i=0}^{n-1} \sum_{j=i}^n (j - i)$$

Brute force approach

- Let's start with the simplest thinkable solution

1. 💡 Algorithmic Idea

try all contiguous subarrays $A[i..j]$

2. </> Pseudocode

```

1  s = 0
2  for i = 0, ..., n - 1
3      for j = i, ..., n
4          t = 0
5          for k = i, ..., j - 1
6              t = t + A[k]
7          end for
8          if t > s then s := t
9      end for
10 end for
    
```

3. ☉ Correctness proof

direct by definition of s

Maximal subarray problem

- **Given:** $A[0..n]$ with $A[i] \in \mathbb{Z}$ for $0 \leq i < n$.
- Abbreviate $s(i, j) := \sum_{k=i}^{j-1} A[k]$
- **Goal:** Compute $s := \max\{s(i, j) : 0 \leq i \leq j \leq n\}$ and a pair (i, j) with $s = s(i, j)$.

4. 🏔️ Algorithm analysis

additions

$$\begin{aligned}
 &= \sum_{i=0}^{n-1} \sum_{j=i}^n \sum_{k=i}^{j-1} 1 = \sum_{i=0}^{n-1} \sum_{j=i}^n (j-i) \\
 &= \sum_{i=0}^{n-1} \sum_{j=0}^{n-i} j = \frac{(n-i)(n-i+1)}{2}
 \end{aligned}$$

$$j=i \leadsto j-i=0$$

$$j=n \leadsto j-i=n-i$$

$$\begin{aligned}
 \sum_{j=1}^m j &= 1 + 2 + 3 + 4 + \dots + (m-1) + m \\
 + \sum_{j=1}^m j &= m + (m-1) + (m-2) + (m-3) + \dots + 2 + 1 \\
 &= (m+1) + (m+1) + (m+1) + (m+1) + \dots + (m+1) + (m+1) \\
 &= m(m+1)
 \end{aligned}$$

Brute force approach

- Let's start with the simplest thinkable solution

1. 💡 Algorithmic Idea

try all contiguous subarrays $A[i..j]$

2. </> Pseudocode

```
1 s = 0
2 for i = 0, ..., n - 1
3     for j = i, ..., n
4         t = 0
5         for k = i, ..., j - 1
6             t = t + A[k]
7         end for
8         if t > s then s := t
9     end for
10 end for
```

3. ☉ Correctness proof

direct by definition of s

Maximal subarray problem

- **Given:** $A[0..n]$ with $A[i] \in \mathbb{Z}$ for $0 \leq i < n$.
- Abbreviate $s(i, j) := \sum_{k=i}^{j-1} A[k]$
- **Goal:** Compute $s := \max\{s(i, j) : 0 \leq i \leq j \leq n\}$ and a pair (i, j) with $s = s(i, j)$.

4. 🏔️ Algorithm analysis

additions

$$\begin{aligned} &= \sum_{i=0}^{n-1} \sum_{j=i}^n \sum_{k=i}^{j-1} 1 = \sum_{i=0}^{n-1} \sum_{j=i}^n (j - i) \\ &= \sum_{i=0}^{n-1} \sum_{j=0}^{n-i} j = \sum_{i=0}^{n-1} \frac{(n-i)(n-i+1)}{2} \end{aligned}$$

Brute force approach

- Let's start with the simplest thinkable solution

1. 💡 Algorithmic Idea

try all contiguous subarrays $A[i..j]$

2. </> Pseudocode

```
1 s = 0
2 for i = 0, ..., n - 1
3     for j = i, ..., n
4         t = 0
5         for k = i, ..., j - 1
6             t = t + A[k]
7         end for
8         if t > s then s := t
9     end for
10 end for
```

3. ☉ Correctness proof

direct by definition of s

Maximal subarray problem

- **Given:** $A[0..n]$ with $A[i] \in \mathbb{Z}$ for $0 \leq i < n$.
- Abbreviate $s(i, j) := \sum_{k=i}^{j-1} A[k]$
- **Goal:** Compute $s := \max\{s(i, j) : 0 \leq i \leq j \leq n\}$ and a pair (i, j) with $s = s(i, j)$.

4. 🏔️ Algorithm analysis

additions

$$\begin{aligned} &= \sum_{i=0}^{n-1} \sum_{j=i}^n \sum_{k=i}^{j-1} 1 = \sum_{i=0}^{n-1} \sum_{j=i}^n (j-i) \quad \begin{array}{l} i=0 \leadsto n-i=n \\ i=n-1 \leadsto n-i=1 \end{array} \\ &= \sum_{i=0}^{n-1} \sum_{j=0}^{n-i} j = \sum_{i=0}^{n-1} \frac{(n-i)(n-i+1)}{2} \\ &= \frac{1}{2} \sum_{i=1}^n i(i+1) = \frac{1}{2} \sum_{i=1}^n i^2 + \frac{1}{2} \sum_{i=1}^n i \end{aligned}$$

Brute force approach

- Let's start with the simplest thinkable solution

1. 💡 Algorithmic Idea

try all contiguous subarrays $A[i..j]$

2. </> Pseudocode

```
1 s = 0
2 for i = 0, ..., n - 1
3     for j = i, ..., n
4         t = 0
5         for k = i, ..., j - 1
6             t = t + A[k]
7         end for
8         if t > s then s := t
9     end for
10 end for
```

3. ☉ Correctness proof

direct by definition of s

Maximal subarray problem

- **Given:** $A[0..n]$ with $A[i] \in \mathbb{Z}$ for $0 \leq i < n$.
- Abbreviate $s(i, j) := \sum_{k=i}^{j-1} A[k]$
- **Goal:** Compute $s := \max\{s(i, j) : 0 \leq i \leq j \leq n\}$ and a pair (i, j) with $s = s(i, j)$.

4. 🏔️ Algorithm analysis

additions

$$\begin{aligned} &= \sum_{i=0}^{n-1} \sum_{j=i}^n \sum_{k=i}^{j-1} 1 = \sum_{i=0}^{n-1} \sum_{j=i}^n (j - i) \\ &= \sum_{i=0}^{n-1} \sum_{j=0}^{n-i} j = \sum_{i=0}^{n-1} \frac{(n-i)(n-i+1)}{2} \\ &= \frac{1}{2} \sum_{i=1}^n i(i+1) = \frac{1}{2} \sum_{i=1}^n i^2 + \frac{1}{2} \sum_{i=1}^n i \\ &= \frac{n(n+1)(2n+1)}{12} + \frac{n(n+1)}{4} \end{aligned}$$

Brute force approach

- Let's start with the simplest thinkable solution

1. 💡 Algorithmic Idea

try all contiguous subarrays $A[i..j]$

2. </> Pseudocode

```
1 s = 0
2 for i = 0, ..., n - 1
3     for j = i, ..., n
4         t = 0
5         for k = i, ..., j - 1
6             t = t + A[k]
7         end for
8         if t > s then s := t
9     end for
10 end for
```

3. ☉ Correctness proof

direct by definition of s

Maximal subarray problem

- **Given:** $A[0..n]$ with $A[i] \in \mathbb{Z}$ for $0 \leq i < n$.
- Abbreviate $s(i, j) := \sum_{k=i}^{j-1} A[k]$
- **Goal:** Compute $s := \max\{s(i, j) : 0 \leq i \leq j \leq n\}$ and a pair (i, j) with $s = s(i, j)$.

4. 🏔️ Algorithm analysis

additions

$$\begin{aligned} &= \sum_{i=0}^{n-1} \sum_{j=i}^n \sum_{k=i}^{j-1} 1 = \sum_{i=0}^{n-1} \sum_{j=i}^n (j - i) \\ &= \sum_{i=0}^{n-1} \sum_{j=0}^{n-i} j = \sum_{i=0}^{n-1} \frac{(n-i)(n-i+1)}{2} \\ &= \frac{1}{2} \sum_{i=1}^n i(i+1) = \frac{1}{2} \sum_{i=1}^n i^2 + \frac{1}{2} \sum_{i=1}^n i \\ &= \frac{n(n+1)(2n+1)}{12} + \frac{n(n+1)}{4} \\ &= \frac{n(n+1)(n+2)}{6} \end{aligned}$$

Brute force approach

- Let's start with the simplest thinkable solution

1. 💡 Algorithmic Idea

try all contiguous subarrays $A[i..j]$

2. </> Pseudocode

```
1 s = 0
2 for i = 0, ..., n - 1
3     for j = i, ..., n
4         t = 0
5         for k = i, ..., j - 1
6             t = t + A[k]
7         end for
8         if t > s then s := t
9     end for
10 end for
```

3. ☉ Correctness proof

direct by definition of s

Maximal subarray problem

- **Given:** $A[0..n]$ with $A[i] \in \mathbb{Z}$ for $0 \leq i < n$.
- Abbreviate $s(i, j) := \sum_{k=i}^{j-1} A[k]$
- **Goal:** Compute $s := \max\{s(i, j) : 0 \leq i \leq j \leq n\}$ and a pair (i, j) with $s = s(i, j)$.

4. 🏔️ Algorithm analysis

additions

$$\begin{aligned} &= \sum_{i=0}^{n-1} \sum_{j=i}^n \sum_{k=i}^{j-1} 1 = \sum_{i=0}^{n-1} \sum_{j=i}^n (j - i) \\ &= \sum_{i=0}^{n-1} \sum_{j=0}^{n-i} j = \sum_{i=0}^{n-1} \frac{(n-i)(n-i+1)}{2} \\ &= \frac{1}{2} \sum_{i=1}^n i(i+1) = \frac{1}{2} \sum_{i=1}^n i^2 + \frac{1}{2} \sum_{i=1}^n i \\ &= \frac{n(n+1)(2n+1)}{12} + \frac{n(n+1)}{4} \\ &= \frac{n(n+1)(n+2)}{6} \sim \frac{1}{6}n^3 = \Theta(n^3) \end{aligned}$$

Reusing sums

1. 💡 Algorithmic Idea

- ▶ brute force algorithm is unnecessarily wasteful!
- ▶ can use $s(i, j) = s(i, j - 1) + A[j - 1]$

Reusing sums

1. 💡 Algorithmic Idea

- ▶ brute force algorithm is unnecessarily wasteful!
- ▶ can use $s(i, j) = s(i, j - 1) + A[j - 1]$

2. </> Pseudocode

```
1 s = 0
2 for i = 0, ..., n - 1
3     t = 0
4     for j = i + 1, ..., n
5         t = t + A[j - 1]
6         if t > s then s := t
7     end for
8 end for
```

3. ☉ Correctness proof: as above

4. 🏔 Algorithm analysis: $\sum_{i=0}^{n-1} \sum_{j=i+1}^n 1 = \frac{n(n+1)}{2} \sim \frac{1}{2}n^2 = \Theta(n^2)$ additions

Reusing sums

1. 💡 Algorithmic Idea

- ▶ brute force algorithm is unnecessarily wasteful!
- ▶ can use $s(i, j) = s(i, j - 1) + A[j - 1]$

2. </> Pseudocode

```
1 s = 0
2 for i = 0, ..., n - 1
3     t = 0
4     for j = i + 1, ..., n
5         t = t + A[j - 1]
6         if t > s then s := t
7     end for
8 end for
```



Can we possibly do better?

- ▶ There are $\binom{n}{2} \sim \frac{1}{2}n^2$ different $s(i, j) \dots$
- $\rightsquigarrow$ Can't look at all of them

3. ☉ Correctness proof: as above

4. 🏔 Algorithm analysis: $\sum_{i=0}^{n-1} \sum_{j=i+1}^n 1 = \frac{n(n+1)}{2} \sim \frac{1}{2}n^2 = \Theta(n^2)$ additions

2.6 Teaser: Algorithm Design in Action

A subquadratic solution

💡 Algorithmic idea:

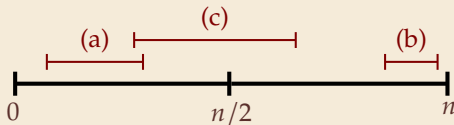
Consider $n/2$ -mark.

Only 3 options for optimal solution $s(i, j)$:

(a) $0 \leq i \leq j < \lceil \frac{n}{2} \rceil$ (left)

(b) $\lceil \frac{n}{2} \rceil \leq i \leq j \leq n$ (right)

(c) $i < \lceil \frac{n}{2} \rceil \leq j$ (straddle)



A subquadratic solution

💡 Algorithmic idea:

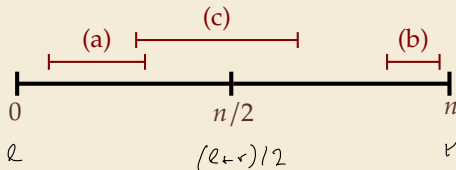
Consider $n/2$ -mark.

Only 3 options for optimal solution $s(i, j)$:

(a) $0 \leq i \leq j < \lceil \frac{n}{2} \rceil$ (left)

(b) $\lceil \frac{n}{2} \rceil \leq i \leq j \leq n$ (right)

(c) $i < \lceil \frac{n}{2} \rceil \leq j$ (straddle)



💡 optimal straddle easy to compute!

- ▶ **independently** find best left endpoint i for $s(i, \lceil \frac{n}{2} \rceil)$ and best right endpoint j for $s(\lceil \frac{n}{2} \rceil, j)$

- ▶ for (a) and (b), recurse on instance of half the size!

A subquadratic solution – Pseudocode & Correctness

```
1 procedure findMaxSubarraySum( $A[\ell..r]$ ):
2   if  $r - \ell \leq 0$ 
3     return 0
4   if  $r - \ell == 1$ 
5     return  $\max\{0, A[\ell]\}$ 
6    $m := \lceil (\ell + r)/2 \rceil$ 
7    $s_{(a)} := \text{findMaxSubarraySum}(A[\ell, m])$ 
8    $s_{(b)} := \text{findMaxSubarraySum}(A[m, r])$ 
9   // Find left endpoint of straddle:
10   $s_\ell := 0; t := 0$ 
11  for  $i = m - 1, m - 2, \dots, \ell$ 
12     $t := A[i] + t$ 
13     $s_\ell := \max\{s_\ell, t\}$ 
14  end for
15  // Find right endpoint of straddle:
16   $s_r := 0; t := 0$ 
17  for  $j = m + 1, \dots, r$ 
18     $t := t + A[j - 1]$ 
19     $s_r := \max\{s_r, t\}$ 
20  end for
21   $s_{(c)} := s_\ell + s_r$ 
22  return  $\max\{s_{(a)}, s_{(b)}, s_{(c)}\}$ 
```

A subquadratic solution – Pseudocode & Correctness

```
1 procedure findMaxSubarraySum( $A[\ell..r]$ ):  
2   if  $r - \ell \leq 0$   
3     return 0  
4   if  $r - \ell == 1$   
5     return  $\max\{0, A[\ell]\}$   
6    $m := \lceil (\ell + r)/2 \rceil$   
7    $s_{(a)} := \text{findMaxSubarraySum}(A[\ell, m])$   
8    $s_{(b)} := \text{findMaxSubarraySum}(A[m, r])$   
9   // Find left endpoint of straddle:  
10   $s_\ell := 0; t := 0$   
11  for  $i = m - 1, m - 2, \dots, \ell$   
12     $t := A[i] + t$   
13     $s_\ell := \max\{s_\ell, t\}$   
14  end for  
15  // Find right endpoint of straddle:  
16   $s_r := 0; t := 0$   
17  for  $j = m + 1, \dots, r$   
18     $t := t + A[j - 1]$   
19     $s_r := \max\{s_r, t\}$   
20  end for  
21   $s_{(c)} := s_\ell + s_r$   
22  return  $\max\{s_{(a)}, s_{(b)}, s_{(c)}\}$ 
```

◎ Correctness proof:

► Induction over $n = r - \ell$

► **basis:** for $n \leq 1$ ✓

► **hypothesis:** Assume
findMaxSubarraySum returns correct
result for all arrays of up to $n - 1$
elements

► **step:** For array of $n \geq 2$ elements,
distinguish cases (a), (b), (c)
(a) and (b) $\rightsquigarrow$ IH ✓
(c) “from inspection of the code”

A subquadratic solution – Analysis

```
1 procedure findMaxSubarraySum( $A[\ell..r]$ ):
2   if  $r - \ell \leq 0$ 
3     return 0
4   if  $r - \ell == 1$ 
5     return  $\max\{0, A[\ell]\}$ 
6    $m := \lceil (\ell + r)/2 \rceil$ 
7    $s_{(a)} := \text{findMaxSubarraySum}(A[\ell, m])$ 
8    $s_{(b)} := \text{findMaxSubarraySum}(A[m, r])$ 
9   // Find left endpoint of straddle:
10   $s_\ell := 0$ ;  $t := 0$ 
11  for  $i = m - 1, m - 2, \dots, \ell$ 
12     $t := A[i] + t$ 
13     $s_\ell := \max\{s_\ell, t\}$ 
14  end for
15  // Find right endpoint of straddle:
16   $s_r := 0$ ;  $t := 0$ 
17  for  $j = m + 1, \dots, r$ 
18     $t := t + A[j - 1]$ 
19     $s_r := \max\{s_r, t\}$ 
20  end for
21   $s_{(c)} := s_\ell + s_r$ 
22  return  $\max\{s_{(a)}, s_{(b)}, s_{(c)}\}$ 
```

 Algorithm analysis:

A subquadratic solution – Analysis

```
1 procedure findMaxSubarraySum( $A[\ell..r]$ ):  
2   if  $r - \ell \leq 0$   
3     return 0  
4   if  $r - \ell == 1$   
5     return  $\max\{0, A[\ell]\}$   
6    $m := \lceil (\ell + r)/2 \rceil$   
7    $s_{(a)} := \text{findMaxSubarraySum}(A[\ell, m])$   
8    $s_{(b)} := \text{findMaxSubarraySum}(A[m, r])$   
9   // Find left endpoint of straddle:  
10   $s_\ell := 0; t := 0$   
11  for  $i = m - 1, m - 2, \dots, \ell$   
12     $t := A[i] + t$   
13     $s_\ell := \max\{s_\ell, t\}$   
14  end for  
15  // Find right endpoint of straddle:  
16   $s_r := 0; t := 0$   
17  for  $j = m + 1, \dots, r$   
18     $t := t + A[j]$   
19     $s_r := \max\{s_r, t\}$   
20  end for  
21   $s_{(c)} := s_\ell + s_r$   
22  return  $\max\{s_{(a)}, s_{(b)}, s_{(c)}\}$ 
```

Algorithm analysis:

- Write $n = r - \ell$
 - # additions in non-recursive part:
 $(m - \ell) + (r - m) + 1 = n + 1$
 - Write $C(n)$ for total # additions for n elements
- $\rightsquigarrow C(n) = C(\lceil \frac{n}{2} \rceil) + C(\lfloor \frac{n}{2} \rfloor) + n + 1$
- for $n = 2^k$ for $k \in \mathbb{N}_0$, this simplifies to
 $C(2^k) = 2C(2^{k-1}) + 2^k + 1$

A subquadratic solution – Analysis

```
1 procedure findMaxSubarraySum( $A[\ell..r]$ ):  
2   if  $r - \ell \leq 0$   
3     return 0  
4   if  $r - \ell == 1$   
5     return  $\max\{0, A[\ell]\}$   
6    $m := \lceil (\ell + r)/2 \rceil$   
7    $s_{(a)} := \text{findMaxSubarraySum}(A[\ell, m])$   
8    $s_{(b)} := \text{findMaxSubarraySum}(A[m, r])$   
9   // Find left endpoint of straddle:  
10   $s_\ell := 0$ ;  $t := 0$   
11  for  $i = m - 1, m - 2, \dots, \ell$   
12     $t := A[i] + t$   
13     $s_\ell := \max\{s_\ell, t\}$   
14  end for  
15  // Find right endpoint of straddle:  
16   $s_r := 0$ ;  $t := 0$   
17  for  $j = m + 1, \dots, r$   
18     $t := t + A[j - 1]$   
19     $s_r := \max\{s_r, t\}$   
20  end for  
21   $s_{(c)} := s_\ell + s_r$   
22  return  $\max\{s_{(a)}, s_{(b)}, s_{(c)}\}$ 
```

Algorithm analysis:

- Write $n = r - \ell$
 - # additions in non-recursive part:
 $(m - \ell) + (r - m) + 1 = n + 1$
 - Write $C(n)$ for total # additions for n elements
- $\rightsquigarrow C(n) = C(\lceil \frac{n}{2} \rceil) + C(\lfloor \frac{n}{2} \rfloor) + n + 1$
- for $n = 2^k$ for $k \in \mathbb{N}_0$, this simplifies to
 $C(2^k) = 2C(2^{k-1}) + 2^k + 1$

$\rightsquigarrow C(n) \sim n \log_2(n)$

A lower bound

- **Theorem:** Every correct algorithm has a running time of $\Omega(n)$.

Beweis: "Wir müssen alle Zahlen lesen."

Adversary argument

Angenommen Alg. $\mathcal{A}$ der nur $o(n)$ Zeit braucht

$\Rightarrow \exists$ Eingabe $A[0..n)$, die $\mathcal{A}$ nicht komplett liest, z.B. $A[i]$ nicht.

Egal was $\mathcal{A}$ ausgibt,

modifizieren von $A[i]$ ergibt Widerspruch.

$\Rightarrow$ Komplexität vom Max Subarray Problem ist $\Omega(n)$ und $O(n \log n)$

An optimal algorithm

💡 **Algorithmic idea:** (dynamic programming)

In a clever sweep, we can compute best $s(i, r)$ and best $s(i, j)$ with $i \leq j \leq r$ for all r .

</> Pseudocode

```
1 procedure findMaxSubarraySum( $A[0..n]$ )
2   suffixMax := 0; globalMax := 0
3   for  $r = 1, \dots, n$ 
4     suffixMax :=  $\max\{\text{suffixMax} + A[r - 1], 0\}$ 
5     globalMax :=  $\max\{\text{globalMax}, \text{suffixMax}\}$ 
6   return globalMax
```

An optimal algorithm

💡 Algorithmic idea:

In a clever sweep, we can compute best $s(i, r)$ and best $s(i, j)$ with $i \leq j \leq r$ for all r .

</> Pseudocode

```
1 procedure findMaxSubarraySum( $A[0..n]$ )
2   suffixMax := 0; globalMax := 0
3   for  $r = 1, \dots, n$ 
4     suffixMax :=  $\max\{\text{suffixMax} + A[r - 1], 0\}$ 
5     globalMax :=  $\max\{\text{globalMax}, \text{suffixMax}\}$ 
6   return globalMax
```

🎯 Correctness proof:

- ▶ Invariant: $\text{globalMax} = \max\{0\} \cup \{A[i] + \dots + A[j - 1] : 0 \leq i < j \leq r\}$ and $\text{suffixMax} = \max\{0\} \cup \{A[i] + \dots + A[r - 1] : 0 \leq i < r\}$
 - ▶ Induction over r

An optimal algorithm

💡 Algorithmic idea:

In a clever sweep, we can compute best $s(i, r)$ and best $s(i, j)$ with $i \leq j \leq r$ for all r .

</> Pseudocode

```
1 procedure findMaxSubarraySum( $A[0..n]$ )  
2   suffixMax := 0; globalMax := 0  
3   for  $r = 1, \dots, n$   
4     suffixMax :=  $\max\{\text{suffixMax} + A[r - 1], 0\}$   
5     globalMax :=  $\max\{\text{globalMax}, \text{suffixMax}\}$   
6   return globalMax
```

🎯 Correctness proof:

- ▶ Invariant: $\text{globalMax} = \max\{0\} \cup \{A[i] + \dots + A[j - 1] : 0 \leq i < j \leq r\}$ and
 $\text{suffixMax} = \max\{0\} \cup \{A[i] + \dots + A[r - 1] : 0 \leq i < r\}$
 - ▶ Induction over r

🏔 Algorithm analysis:

- ▶ clearly n additions